

The Origins of Tense-based Case Marking in Pitta-Pitta and Wangkajutjuru¹

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Several Australian languages show tense marking on non-verbal items, including the Karnic languages Pitta-Pitta and Wangkajutjuru (formerly spoken in Western Queensland). In these languages the forms of core case markers and the ergative/accusative split vary according to the tense and mood of the clause. Here I describe tense marking in the languages and the differences between sources and dialects, and discuss the etymologies of the case markers. I discuss sources for the future-marking construction and argue that these tense-marking patterns arose from the reanalysis of a construction marking reduced transitivity. While several other constructions have been proposed in the literature as the source for Pitta-Pitta's case marking, I argue here that they do not adequately account for all the facts.

1. Introduction

Several Australian languages have come to the notice of syntacticians in recent years because of their unusual tense marking strategies (see, for example, Hale 1998; Pesetsky & Torrego 2001). We find in these languages a number of ways in which tense may be marked in a clause, not all of these involving the verb. In Kayardild (Evans 1995), for example, tense can be marked on all non-subject arguments; the tense is signalled by the case used. In Baagandji (Hercus 1982), however, non-verbal tense marking appears only on subject pronouns. The situation is different again in the Karnic language Pitta-Pitta (Roth 1897; Blake & Breen 1971; Blake 1979c), where tense is marked on both subject and (depending on the dialect) object. This paper focuses on Pitta-Pitta and the very closely related Wangkajutjuru, and the historical origin of their tense-determined case marking strategies.

The first part of the paper is a description of Pitta-Pitta's tense marking (verbal and non-verbal). In the remainder of the paper I will examine the comparative/historical evidence for the etymology of the construction and draw some parallels with other Pama-Nyungan languages. There are at least three possible origins for this type of case marking in Pitta-Pitta. Two of them are a reanalysed anti-passive

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[suggested by Hopper and Thomson (1980) and Dixon (2002) for Pitta-Pitta] and ‘insubordination’ [as suggested by Evans (1995) for Kayardild]. I argue here that neither of these is the origin of Pitta-Pitta’s tense marking strategy, which is instead parallel historically to the ‘reduced transitivity’ (or conative) construction of some other Pama-Nyungan languages, such as Jalarnnga, Kalkatungu and Warlpiri. Only the last of the alternative etymologies can satisfactorily account for the patterns we find in the modern languages.

1.1. The Languages

Pitta-Pitta is a member of the Karnic subgroup of Pama-Nyungan (Breen 1971; Bovern 2001a), the northern-most language of the subgroup, formerly spoken in the area around Boulia in far Western Queensland. There are no longer any full speakers of this language. The linguistic situation of the Boulia district was quite complicated, and Pitta-Pitta was in fact only one of a number of dialects spoken in the area. These dialects fall into two groups. The Eastern group comprised Pitta-Pitta and a number of very badly attested other dialects, the names of which are Rangwa, Kunkalanya, Ngulupulu and Ringa-Ringa; the Western group was made up of Wangkajutjuru and Lhanima. The languages are all closely related, and the Eastern and Western groups seem to have formed a pair of dialect continua. Here I use Pitta-Pitta as the representative dialect of the Eastern languages, and Wangkajutjuru as the representative of the Western languages, since they are the best described.

My data for Pitta-Pitta and Wangkajutjuru are from Blake and Breen (1971), Blake (1979c) and Roth (1897). Blake (1979c) and Blake and Breen (1971) are based on the same material; Roth (1897) was written after its author’s travels in Western Queensland as Protector of Aborigines. Blake and Breen (1971) is a presentation and preliminary analysis of fieldwork conducted by Gavan Breen and Barry Blake in the mid to late 1960s, primarily with Mrs Nora Jacks. Blake (1979c) is a sketch, based on the same material, with analyses which draw different conclusions from Blake and Breen (1971) in some cases.² Roth (1897) is a reasonable summary of the main features of the language, as far as it goes. The phonology is very detailed but also rather inaccurate. Roth does give many example sentences, however, including fairly complicated multiclausal constructions. His data also show less interference from English than later recordings—while Roth’s informants would have still spoken Pitta-Pitta amongst themselves, Blake’s and Breen’s informants all used English as their primary language of communication.

2. Synchronic Analysis of Tense Marking

In this section I give a synchronic analysis of the facts of case and tense marking in Pitta-Pitta and Wangkajutjuru.

² Gavan Breen also shared some field transcriptions with Mrs Nora Jacks in 1970. Examples from this additional material are referenced as Breen (fieldnotes).

2.1. The Forms

In Pitta-Pitta and Wangkajutjuru, the forms of case markers and the ergative/nominative/accusative case split depends on the tense of the clause. The split distinguishes between ‘future’ contexts and ‘non-future’ ones, where ‘future’ contexts include the future tense, the potential and purposive modalities (and their equivalent in subordinate clauses) and imperatives.³ ‘Non-futures’ are the other tenses and moods.

By way of an illustration of Pitta-Pitta non-future and future clauses and their differences, consider the following two examples from Roth (1897), our oldest data for Pitta-Pitta.⁴

- | | | | |
|-----|--|--|--|
| (1) | Kanari-pakoola- <u>lo</u>
<i>kiniyari-parrkula-lu</i>
young woman-DL-NFUT.ERG
‘Two young girls were grinding seed.’ | pappa- <u>na</u>
<i>papa-nha</i>
grass seed-NFUT.ACC
grind-PST
(Roth 1897: 12) | wire-ka.
<i>wira-ka</i>
grind-PST
(Roth 1897: 12) |
| (2) | Koopa-koopa- <u>ngu</u>
<i>kupakupa-ngu</i>
old man-FUT.E/N
‘The old man will steal the spears.’ | biba-pooro-pityiri- <u>ko</u>
<i>pipapuru-pityiri-ku</i>
spear-PL-FUT.ACC
(Roth 1897: 14) | punnare.
<i>punhara(?) -o</i>
steal-FUT
(Roth 1897: 14) |

In example (1), the verb is marked for past tense, the ergative case is *-lu* and the accusative *-nha*. In the future tense-marked example (2), however, the ergative is *-ngu* and the accusative is *-ku*. A look at intransitive clauses shows that nominative

³ Because sentences in the future tense are by far the most common instantiation of the marking pattern, I will continue to refer to the allomorphs as the ‘future’ allomorphs, even though they occur in other frames as well. There are also a few exceptions, which are discussed in Section 2.2 below. Note that ‘future contexts’ are defined by verb morphology rather than on purely semantic grounds, as there are sentences which refer to future time but have present verb morphology, non-future case morphology and a dubitative particle. Such examples are not further discussed here.

⁴ Abbreviations used in this paper are: ABL, ablative; ABS, absolutive; ACC, accusative; ALL, allative; ANTI.PASS, antipassive; BEN, benefactive; COMIT, comitative; COMP, complementizer; COND, conditional; CONT, continuous; DAT, dative; DEIX, marker of general deixis; DIST, distal deixis; DL, dual; ERG, ergative; FEM, feminine; FUT, future; FUT.E/N, future ergative/nominative; GEN, genitive; GER, gerund; IMP, imperfective; IMPER, imperative; INST, instrumental; LOC, locative; M.LOC, modal locative; M.PROP, modal proprietive; NFUT, non-future; NEG, negative; NOM, nominative; OBJ, object; PL, plural; POSS'D, possessed; POT, potential; PP, Pitta-Pitta; PRES, present; PROX, proximal deixis; PST, past; PURP, purposive; RECIP, reciprocal; SG, singular; TRANS, transitive; TRANSLOC, translocative; UNDEF, undefined deixis; WJ, Wangkajutjuru. In the orthography, *R* represents a retroflex glide, *r* a tap and *rr* a trill; *ng* is a velar nasal. *h* following a nasal or stop indicates that the stop is laminodental. In examples from Roth (1897), the first line is Roth’s transcription of the sentences. The second line is my conversion of Roth’s transcription to phonemic transcription in standard Karnic orthography (for example, Hercus 1994). A question mark in brackets in that line means that the word was not recorded by Blake and Breen (1971) and phonemicization is ambiguous. Morphemes are separated by hyphens in all lines of the examples. Portmanteaux morphemes that are not segmentable are glossed with parts separated by full stops (e.g. *nganyu*, 1SG.FUT.E/N, the first person singular future ergative/nominative). Unless otherwise noted, all examples are from Pitta-Pitta.

arguments are also marked differently in future and non-future tenses, as examples (3) and (4) below show:

- (3) Wallara machoomba punnia.
walara(?)-*ø* *matyumpa*-*ø* *panha*-*ya*
 young-NFUT.NOM kangaroo-NFUT.NOM run-PRES
 ‘A young kangaroo is running.’
- (4) Munna-*ngo* wungata-pityiri-*ngo* mullo nungka.
manha-*ngu* *wangarta(?)*-pityiri-*ngu* *malhu* *nhangka*-*ø*
 bad-FUT.E/N woman-PL-FUT.E/N NEG sit-FUT
 ‘The bad women will not sit down.’

Intransitive non-future subjects receive a null case marker. This is shown in example (3), on *walara matyumpa* ‘young kangaroo’. Intransitive future subjects are marked the same way as transitive future subjects, that is, with *-ngu*, as in example (4).⁵ A full paradigm of the forms found in Roth (1897) is given in Table 1 below.

From the data we can see that nominals in Pitta-Pitta and Wangkajutjuru are marked according to a nominative/accusative pattern in the future, while in other tenses an ergative/nominative/accusative is in evidence.

In the data in Blake and Breen (1971) and Blake (1979c) the same case forms are found, although there are some differences in their usage (to be discussed below). Some examples are given in (5) and (6) below:

- (5) Examples in past tense:
- a. Warra-*ø* *nha*-*para* *ngura*-*minya* *karnrta*-*nya*. (WJ)
 woman-NFUT.NOM 3SG.FEM.NFUT.NOM-DEIX camp-ABL go-PST
 ‘The woman went away from the camp.’
- b. WaiRa-*nha* *ngathu* *thari*-*nya* *nhana*-*yi*
 heart-(NFUT.)ACC 1SG.NFUT.ERG spear-PST 3SG.(NFUT.)ACC-PROX
wakira-*nha*. (WJ)
 kangaroo-(NFUT.)ACC
 ‘I speared the kangaroo in the heart.’ (Blake & Breen 1971: 90)
- (6) Examples in future contexts:
- a. Yin-*ngu* *pani* *nganupali*-*nha*.
 2SG-FUT.E/N not enter-IMPER
 ‘Don’t you go in.’ (Breen fieldnotes)

⁵ There is a terminological problem in that ‘nominative’ can refer both to subjects of transitive and intransitive clauses when contrasted with ‘accusative’. Here I restrict the term ‘nominative’ to refer only to the subject of an intransitive verb. I use ‘ergative’ to refer to the case assigned to the subject of a transitive verb in non-future context and use ‘ergative/nominative’ to refer to the formal marker of subjects of transitive and intransitive verbs in future contexts (that is, the arguments that can be marked with *-ngu*). Thus I assume a syncretism of ergative and nominative in the future. I also use the term ‘subject’ to refer to the grammatical relation marked by nominative or ergative case. Note that Pitta-Pitta and Wangkajutjuru have no ‘absolutive’ case, since at no point are the subjects of intransitive clauses and the objects of transitive clauses marked the same way.

Table 1. Roth's core case markers for Pitta-Pitta

Gloss	Roth's form	Phonemicization
Ergative (future)	<i>-ngo</i>	<i>-ngu</i>
Nominative (future)	<i>-ngo</i>	<i>-ngu</i>
Ergative (non-future)	<i>-lo</i>	<i>-lu</i>
Nominative (non-future)	<i>[unmarked]</i>	<i>-ø</i>
Accusative (future)	<i>-ko</i>	<i>-ku</i>
Accusative (non-future)	<i>-na</i>	<i>-nha</i>

- b. *Ngapi-ri-ngu tyilatyi-la-ngu thawa-ø paya-nha.* (Lhanima)
 father-FUT.E/N boomerang-FUT.INST kill-FUT bird-NFUT.ACC
 'Father will kill some birds with a boomerang.' (Blake & Breen 1971: 133)
- c. *Tyira-maru-ngu nganyu karnta-ø.*
 boomerang-PROP-FUT.E/N 1SG.FUT.E/N go-FUT
 'I'll be going with the boomerang.' (Blake & Breen 1971: 89)
- d. *Yun-ngu nhana-para thatyi-ya, ira-lhi-ra yun-ngu.* (WJ)
 2SG-FUT.E/N 3SG.ACC-DEIX eat-COND, sick-POT 2SG-FUT.E/N
 'If you eat that [meat], you might get sick.' (Blake & Breen 1971: 133)

The sentences in examples (5) above show transitive and intransitive clauses with the non-future allomorphs of the structural case markers. Example (5a) is intransitive; the single argument is marked for non-future nominative case (marked by *-ø*). In (5b) the sentence is transitive; the subject is marked with the non-future ergative *-lu* and the constituents of the discontinuous object NP are marked with the accusative *-nha*. In the future tense-marked sentences in (6a–d), we find *-ngu* for 'subjects', whether ergative or nominative [just as in Roth (1897)]. The accusative-marked arguments, however, show *-nha*, rather than the *-ku* expected from Roth's data.

2.2. Differences between Dialects and Sources

There are some dialectal differences between Pitta-Pitta and Wangkajutjuru, as well as between the Pitta-Pitta recorded by Blake and Breen (1971) and that recorded by Roth (1897).

In Pitta-Pitta both subject and object cases vary for tense, whereas in Wangkajutjuru only the subject allomorphs vary; in Wangkajutjuru all objects, whether in future or non-future clauses, take *-nha* (or the equivalent accusative allomorphs in pronouns). This is illustrated by example (6d) above, for example, where the third person pronoun is *nhana-para*, and not *nhaku-para*, with the future allomorph of the accusative. A further example is given in (7) below. Note the marking of the subject pronoun with future *-ngu* and object noun with *-nha*.⁶

⁶ Note that Roth (1897) gives data only for Pitta-Pitta so we have no information as to whether Wangkajutjuru's allomorphy conditions were present 100 years ago and represent a dialectal difference or are a result of language death.

- (7) *Nhu-ngu-para pirta kawaRa-nha.* (WJ)
 3SG-FUT.E/N-GEN hit-FUT dog-(FUT).ACC
 ‘He will hit the dog.’ (Blake & Breen 1971: 140)

Another difference in the corpora is the use of the future accusative *-ku* in Pitta-Pitta. In Roth’s Pitta-Pitta data, all future objects are marked with *-ku* (Roth 1897: 3, 8). In Blake and Breen’s corpus, however, *-ku* is optional, and *-nha* frequently appears where we would expect future objects marked by *-ku*. It seems that *-ku* is most frequently used in the corpus with pronouns (Blake & Breen 1971: 89–91). Blake and Breen’s Pitta-Pitta informant accepted the use of *-ku* on nouns, however, so it is probably the case that the Roth data is closest to ‘correct’ Pitta-Pitta and the optional use of *-nha* in the future is related to the moribund state of the language.⁷

A final point to note is that in Blake and Breen’s corpus, there occasionally appear alternations in future subject case marking. Compare examples (8a) and (8b) below:

- (8)
- a. *Wama-ngu nhu-ngu-para patya-lhiRa.* (WJ)
 snake-FUT.E/N that-FUT.E/N-DIST bite-POT
 ‘That snake might bit you.’
- b. *Tyira-lu ngathu ina pita-lhiRa.* (Lhanima)
 boomerang-NFUT.INST 1SG.NFUT.ERG 2SG.(NFUT).ACC hit-POT
 ‘I might hit you with a boomerang.’ (Blake & Breen 1971: 114)

Such examples are rare in the published corpora; they appear only in Blake and Breen (1971). Example (8b) is doubly unusual since we would expect the ergative of *tyira* in Lhanima to be *tyira-ru*, not **tyira-lu* [although Blake and Breen (1971: 114) note that the few examples of *-lu* for *-ru* in Lhanima are after stems containing *r*, so it may be the result of dissimilation]. Note also that most of the cases where future and non-future marking alternate are with the potential and purposive inflections *-liRa* and *-linga*. The variable uses of non-future case allomorphy could be related to the moribund state of the language. Alternatively, it could be that the optionality is indicative of a change in progress.

2.3. Tense-based Allomorphy of the Instrumental

Another point to note is that in the data in both Blake and Breen (1971) and Roth (1897), the instrumental case also participates in tense-based allomorphy. The instrumental case suffixes are identical in form and patterning to the ergative forms, with *-lu* appearing in the non-future tenses and *-ngu* in future contexts. An example showing an instrumental noun phrase in the future was given in (6b) above. Three further examples are given below. Examples (9a) and (9b) are from Roth’s data; (9a) is in the present tense with the instrumental marked by *-lu* (note also *murawa-lu* ‘woman-NFUT.ERG’, the subject of the clause), while (9b) and (9c) are in the future.

⁷ Alternatively, the optionality of the use of *-ku* on objects in future contexts may be evidence for a syntactic change in progress.

(9)

- a. Moyerjo-lo wakerdi-na miriwinni-lo touea.
muruwa-lu wakirti-nha miriwinhi(?) -lu thawa-ya
 woman-NFUT.ERG crow-NFUT.ACC stone-NFUT.INST kill-PRES
 'The woman is killing the crow with a stone.' (Roth 1897: 10)
- b. Mungkora-ko kope-ko mali-ngo kache nunyo.
mangkura(?) -ku kupi-ku marli-ngu katya-ø nganyu
 much-FUT.ACC fish-FUT.ACC net-FUT.INST catch-FUT 1SG.FUT.E/N
 'I shall catch a large quantity of fish with my net.' (Roth 1897: 27)
- c. *MaRa-ngu nganhu pirta-linga.* (WJ)
 hand-FUT.INST 1SG.FUT.E/N hit-PURP
 'I'm going to hit him with my hand.' (Blake & Breen 1971: 89)

It is not surprising that the instrumental case participates in tense-based case marking along with the ergative; in the majority of Pama-Nyungan the ergative and the instrumental are homophonous [see, for example, Blake (1977) and Austin (1981: 48–49, 119) for examples from the Karnic language Diyari].

2.4. Uses of *-ku* in Present Tense

While the morpheme *-ku* is used with direct objects in future contexts, it is also used in some contexts in the present. In Pitta-Pitta a few verbs regularly subcategorize for indirect objects marked by *-ku*, whatever the tense of the clause. These verbs include: *wapa-* 'look for', *rtiwa-* 'to be jealous of', *yatha-* 'to like'⁸ and *wantili-* 'to wait for'. Examples are given in (10), in the present tense. Note that *-ku* is glossed here as 'dative'.

- (10) a. *Yatha-ya nganytya in-ku.*
 like-PRES 1SG.NFUT.NOM 2SG-DAT
 'I like you.' (Blake 1979c: 197)
- b. *Nganytya muyutyu-ku rtiwa-ya.*
 1SG.NFUT.NOM old woman-DAT be jealous-PRES
 'I'm jealous of the old woman.' (Blake 1979a: 334)

Note that these verbs are structurally intransitive: they take nominative subjects in non-future environments and must be transitivized in order to take reflexive morphology, as example (11) below shows:

- (11) *Rtiwa-la-mali-ya marna pula-ka.*
 jealous-TRANS-RECIP-PRES might 3DL.NFUT.NOM-DEIX
 'They might be jealous of one another.' (Blake 1979a: 334)

⁸ Note that this verb was also recorded with accusative marking in Blake and Breen (1971: 105) (and as *yarta*, rather than *yatha*), although note the nominative, not ergative subject, so the verb is not transitive.

(i) *Yarta-ya nganytya inha-ka.*
 like-PRES 1SG.NFUT.NOM 3SG.NFUT.ACC-DEIX
 I like him.'

The status of such verbs in Wangkajutjuru is difficult to determine. The verb *wapa-* ‘look for’ appears to take a benefactive object, as does the verb *yatha-* ‘to like’ (Blake & Breen 1971: 132). The benefactive is marked by *-nga* or *-yanga* (on pronouns). An example is given in (12):

- (12) *Wapa-nga-nya nginytya yinka-yanga.*
 look for-IMP-PAST 1SG.NFUT.NOM 2SG.DAT-BEN
 ‘I looked for you.’

This marking perhaps implies that the functions of *-ku* are being eroded in the language, being replaced either by *-nga* or by *-nha* in different contexts.

Another context in which *-ku* appears in non-future contexts is to mark motion across a place, as shown in example (13).

- (13) *Yurta-ka nhuwa-ka ngarraRu-ku.*
 swim-PST 3SG.NFUT.NOM-PROX flood-TRANSLOC
 ‘He swam across the floodwaters.’ (Blake 1979c: 197)

Such examples could probably be subsumed under the indirect objects marked by *-ku* [as illustrated for example in (10) above]. Roth (1897: 13) gives a similar example of *-ku* in translocative function but it is in the future tense and cannot be clearly distinguished from regular future marking.

Finally, *-ku* marks the demoted object of anti-passives; an example is given in (14a) below. Example (14b) shows the corresponding underived transitive construction. The anti-passive in Pitta-Pitta is used primarily to mark an intention. Note the morphology of the construction: the subject is nominative, the verb is marked with *-li* and the object is marked by *-ku*.

- (14) a. *Nganytya thatyi-li-ya kathi-ku.*
 1SG.NFUT.NOM eat-ANTI.PASS-PRES meat-DAT
 ‘I want to have a feed of the meat.’ (Blake 1979c: 207)
 b. *Ngathu thatyi-ka i-nha-ka kathi-nha.*
 1SG.NFUT.ERG eat-PST 3SG-NFUT.ACC-DEIX meat-NFUT.ACC.
 ‘I ate the meat.’

No examples of antipassives in Wangkajutjuru seem to have been published, although Blake and Breen (1971) and Blake (1979c), however, both write that Wangkajutjuru also uses *-ku* to mark antipassives.

2.5. A Note on Verb Morphology

To get a proper picture of the context of case marking, one should also consider verb morphology in Pitta-Pitta and Wangkajutjuru. Table 2 below gives the tense, aspect, modality and derivational suffixes in the two languages (from Blake & Breen 1971; Blake 1979c). Note that there is no zero derivation in Pitta-Pitta and Wangkajutjuru; changes between nominal and verbal word classes are achieved through overt suffixation.

Table 2. Verbal tense, aspect and mood suffixes in Pitta-Pitta and Wangkajutjuru

	Tense	Pitta-Pitta	Wangkajutjuru
Non-future	Present	<i>-ya</i>	<i>-ngi</i> <i>-mi</i>
	Imperfective		<i>-nga</i>
	Past	<i>-ka</i>	<i>-nya</i>
Future	Future	<i>-ø</i>	<i>-ø</i>
	Imperative	<i>-nha</i>	<i>-nha</i> <i>-Ru</i>
	Potential/Lest	<i>-lia</i> <i>-ka(r)la</i>	<i>-liRa</i>
	Conditional		<i>-ya</i>
Nominalized forms	Gerund	<i>-l(h)i</i>	<i>-l(h)i</i>
	Purposive	<i>-linga</i>	<i>-linga</i>
Other derivational	Anti-passive	<i>-li</i>	<i>-ri</i>
	Reflexive/Reciprocal	<i>-mali</i>	<i>-mari</i>
	Causative	<i>-Ri</i>	<i>-la</i>
	Transitivizer	<i>-kuri</i>	<i>-kuri</i>
	Detransitivizer	<i>-kana</i>	<i>-kana</i>

2.6. Subordinate Clauses

In this section I give an outline of the formation of subordinate clauses in the Eastern and Western language, since this will be relevant to arguments concerning the origins of future-marking case constructions in Section 4.

The distinction in case marking between the future and non-future also applies in subordinate clauses. Subordinate clauses in Pitta-Pitta are marked by T-complementizing and C-complementizing case [to use the terminology of Dench and Evans (1988)]. The verb of the subordinate clause receives a case marker. Which case is used depends on the type of subordinate clause. This frequently results in case stacking; that is, constituents in the clause being marked with more than one case.

Relative clauses and clauses denoting secondary predication receive the case of the co-referential argument in the matrix clause. This is Dench and Evan's C-complementizing case. In example (15) below, for example, the subject of *yaringanya* '(were) talk(ing)' is *muruwatyinha*, 'the old ladies-NFUT.ACC'. *Muruwatyinha* is also the object of the matrix verb, *pangkalthitha* 'used to hear', and as such it is marked with non-future accusative case *-nha*. The verb of the subordinate clause is also case-marked with accusative case. Example (15) gives the complex construction and (16) the two simplex clauses it comprises.⁹ Example (17) is another example showing the same phenomenon, with relativization on the matrix clause subject.

⁹ The gerund formative *-l(h)i* together with the third person marker of a possessed noun *-tha* are used to signal the past habitual. The construction is not mentioned in Blake (1979c) and examples are given but the forms are not discussed in Blake and Breen (1971).

- (15) *Ngathu pangka-lhi-tha muruwa-tyi-nha yari-nga-nya-nha.*
 1SG.NFUT.ERG hear-GER-POSS'D old lady-PL-NFUT.ACC talk-IMP-PST-NFUT.ACC
 'I used to hear the old ladies talking.'
 (Lhanima)
 (Blake & Breen 1971: 139)
- (16) a. *Ngathu pangka-lhi-tha.*
 1SG.NFUT.ERG hear-GER-POSS'D
 'I used to hear it.'
 b. *Muruwa-tyi yari-nga-nya.*
 old lady-PL talk-IMP-PST
 'The old ladies were talking.'
- (17) *Nha-para pulpu nhangka-ngi-ngu pirta kawara-nha.* (Lhanima)
 3SG.NFUT.NOM-DEIX far sit-PRES-FUT.E/N hit-FUT dog-(FUT).ACC
 'That one (who is) sitting over there will hit the dog.'
 (Blake & Breen 1971: 140)

The structure I am assuming for such relative clauses is given in (18) below (using a relativized object as an example).

- (18) [... V [NP_i [_i V (NP)_{obj}]_{IP}]NP-OBJ]_{IP}

Temporal and subordinate clauses are also marked by case. They are marked with one of the local cases (such as the ablative *-minya*, the purposive *-nga* or the comitative *-maru*). The verb may also be marked with the case of the co-referential argument, if there is one. This is illustrated by the examples in (19).

- (19) a. *Thatyi-ka-inya-o mutyi-ka nhari ngantya.*
 eat-PST-ABL-NFUT.NOM sleep-PST now 1SG.NFUT.NOM
 'After I ate, I had a sleep.' (Blake & Breen 1971: 141)
 b. *Ngathu thala pithi-ka nhatyi-ka-inya-lu*
 1SG.NFUT.ERG COND hit-PST see-PST-ABL-NFUT.ERG
 'I would have hit him if I'd seen him.' (Blake & Breen 1971: 141)
 c. *Malytyarra nganytya mutyi-ka-maru.*
 fresh 1SG.NFUT.NOM sleep-PST-COMIT
 'I'm fit now that I've had a sleep.' (Blake 1979c: 217)

Some clauses marked with C-complementizing case have case marked on all arguments, while in other examples case is marked only on the verb.

2.7. Summary

We can draw the schema in Table 3 for future and non-future basic clauses in Pitta-Pitta and Wangkajutjuru.

We can summarize the following synchronic and diachronic facts about tense marking in Pitta-Pitta and Wangkajutjuru. The morpheme *-ngu* has no function other than to mark future nominative, ergative or instrumental arguments. The future accusative *-ku* is used in Roth's Pitta-Pitta data as the allomorph of the accusative used in the future tenses; in Blake and Breen (1971) it is used optionally

Table 3. Summary of tense-marking in basic clauses in Pitta-Pitta and Wangkajutjuru

	Tense	Tense/Transitivity	Subject	Object
PP	Non-future	Intrans	<i>-o</i>	
		Trans	<i>-lu</i>	<i>-nha</i>
	Future	Intrans	<i>-ngu</i>	
		Trans		<i>-ku</i>
WJ	Non-future	Intrans	<i>-o</i>	
		Trans	<i>-ru</i>	<i>-nha</i>
	Future	Intrans	<i>-ngu</i>	
		Trans		<i>-nga</i>

and primarily with pronouns; in Wangkajutjuru it is not used at all. In Pitta-Pitta a few verbs take indirect objects marked by *-ku* whatever the tense, such as *wapa* ‘look for’. The allomorphs *-ngu* and (optionally) *-ku* are used in the ‘future’ tense, potential and purposive modalities (and their equivalent in subordinate clauses) and with imperatives.

3. Morphological Reconstruction: Etymologies of Case Markers

In this section I give arguments for reconstructing Pitta-Pitta/Wangkajutjuru *-lu* and *-ngu* as reflexes of old ergatives. Discussion is provided on the etymology of the Pitta-Pitta future object suffix *-ku* and the fate of *-ku* in other Karnic languages.

3.1. ‘Subject’ Cases: *-ngu* and *-lu*

Both *-ngu* and *-lu* can be reconstructed as ergative suffixes. The future ergative/nominative *-ngu* is the regular reflex of the ergative suffix reconstructed for Proto Karnic as **-ngu* and for Proto Pama-Nyungan (the ancestor language of Proto Karnic and a number of other subgroups) as **-ngku*. For the Proto Pama-Nyungan reconstruction compare Blake (1988), Dixon (1980) and Hale (1976). The Proto Karnic sound change **-NCV> -NV*¹⁰ is regular in this environment, although examples are rare (compare also, for example, the reflex of the Proto Pama-Nyungan locative **-ngka*, realized as *-nga*, and Pitta-Pitta’s locative case *-ina*, probably from **-Cinta*). Other Karnic languages which show a reflex of *-ngu* for the ergative are Wangkumara and Garlali (Breen nd; McDonald & Wurm 1979).

The form *-lu* is also widespread in Karnic as an ergative suffix; it is found, for example, in Yandruwandha (Breen 1975), in Wangkumara on masculine nouns (Breen nd), and with the variant *-li* in Diyari, Ngamini and Yarluyandi (Austin 1981). The correspondence *l:r* in the ergative between Pitta-Pitta and Wangkajutjuru is regular. In Arabana-Wangkangurru *l* undergoes pre-stopping when it occurs after a stressed syllable and becomes a tap *r* elsewhere (*l* does not occur word

¹⁰ That is, the simplification of a homorganic nasal-stop cluster to the corresponding nasal.

initially). In Wangkajutjuru there is no prestopping in a post-stress syllable (c.f. *pūli* ‘dry’) but the rule of $l > r$ does apply as in Arabana-Wangkangurru.¹¹

Many Pama-Nyungan languages have two ergative markers, *-ng(k)u* and *-lu*, although the conditioning of the allomorphy varies from language to language (Sands 1996: 9). The environment conditioning *-ngu* and *-lu* was probably syllable-based, or similar. If that was the conditioning environment for Proto-Karnic, it is not directly recoverable, since Proto-Karnic seems to have allowed monosyllabic words, although the daughter languages do not (see further Bowern 1998: Ch. 2). We see different reanalyses of *-ngV* ~ *-IV* morphemes in other Karnic languages, indicating that the conditioning environment has changed between Proto Karnic and the daughter languages, and the modern languages have independently recreated different conditions for allomorphy.

There are one or two other possible candidates for the etymology of *-ngu*, including the Proto Karnic ablative **-ngu*, < Proto Pama-Nyungan **-ngu* + ¹² (Koch 1997). This is a possible match formally, however it is implausible that *-ngu* reflects a former ablative, for several reasons. Firstly, nowhere to my knowledge in the Pama-Nyungan area do we find ablatives marking subjects in any common construction (although ablative subjects are found in non-Pama-Nyungan languages such as Wambaya). There are no ablative subjects in Karnic languages. Karnic languages have no quirky case-marked subjects.¹³ Secondly, we have a ready source of the alternation *-ngu* ~ *-lu* in Pitta-Pitta subject marking if we assume that the forms descend from ergative markers, since a morphological alternation *-ng(k)u* ~ *-lu* is attested in many Pama-Nyungan areas. Thus if we reconstruct Pitta-Pitta’s *-ngu* as a reflex of the ergative, we need only assume that Pitta-Pitta exapted the *-ng(k)u* ~ *-lu* alternation and used it in the tense-based base marking split. If we assume that the future subject morpheme is historically related to the ablative, rather than the ergative, we must assume a considerably more complicated scenario. Finally, if we reconstruct *-ngu* as the reflex of an ergative marker, we have a source for the participation of the instrumental in tense-based case marking, since instrumentals are often homophonous with ergative case forms.

3.2. Object Cases *-nha* and *-ku*

The Pitta-Pitta future accusative form *-ku* is clearly a reflex of the Proto Pama-Nyungan and Proto Karnic nominal dative suffix **-ku*. Reconstructing the future accusative from a former dative allows us to tie in the use of *-ku* to mark the goals of anti-passives and indirect objects such as *wapa-* ‘look for’. We need simply assume that these functions reflect retentions of the function of the Proto-Karnic dative that have been fossilized when other functions of Proto-Karnic *-ku* (such as possession) were taken over by *-nga*.

¹¹ For further details see Bowern (1998: 38ff.).

¹² That is, **-ngu* plus some morphological increment.

¹³ See also Austin (1981: 223ff) for examples from Diyari similar to the ones from Pitta-Pitta and Wangkajutjuru.

Reconstruction within Karnic shows that there has been a shift in the use of *-ku* in Pitta-Pitta and all other Karnic languages away from the primary functions of **-ku* to mark categories such as the purposive. I retain the label ‘dative’ in this discussion, following Blake (1979c); however this is probably not the most accurate label for the morpheme in a synchronic description of Pitta-Pitta morphology. The reflex of the Proto Karnic locative, **-nga*, is used for most functions usually denoted by ‘dative’ in Australian languages, including marking purposives, benefactives and possessives. In contrast, *-ku* is used to mark the demoted object in anti-passive constructions, in the future to mark the object of the verb, and to mark the path of motion in examples such as (13) above.

The non-future accusative *-nha* can be reconstructed to Proto Karnic as *-nha* in the pronominal paradigm, but in Pitta-Pitta and WJ *-nha* has been extended from the pronominal paradigm to mark accusative case on all nouns. Most other Karnic languages show *-nha* on pronouns only, and have a null-marked accusative on nouns (such as Yandruwandha: see further Breen 1975).

3.3. Other Nominal Morphology

As mentioned in the previous section, the Pitta-Pitta purposive *-nga*, which fills many of the functions that the ‘dative’ *-ku* fills in many other Pama-Nyungan languages, is a reflex of the Proto-Karnic locative **-nga*, and the Proto-Pama-Nyungan locative **-ngga*. The shift from Proto-Karnic ‘locative’ to modern ‘dative’ has occurred in almost all Karnic languages, except Arabana-Wangkangurru. Punthamara and Garlali (Holmer 1988; Breen 1976/7) have *-nga*, for example, and Wangkumara (McDonald & Wurm 1979) has *-anga*, built on a reduced pronominal stem. Yawarrawarrka and Yandruwandha (Breen 1975) have *-ngari*, probably based on *-nga* plus an increment.

The Karnic daughter languages which shifted the Proto-Karnic locative to dative functions have innovated new locatives, many of which seem to be traceable either to suffixed deictic particles (e.g. Yandruwandha’s locative *-yi*) or to cliticized inflecting elements, such as pronouns or particles.¹⁴ I argue in Bower (1998: 83ff) that Pitta-Pitta’s local cases can be reconstructed to case-inflecting locational words. The local cases (ablative, locative and allative) appear to be based on a stem *-yi* or *-ma*, both of which can probably be reconstructed back to Proto-Karnic.

3.4. Reconstruction of Verb Morphology

Very little verb morphology can be reconstructed at present to Proto Karnic. This is partly a result of the loss of verb conjugation markers in Karnic languages; it appears that the daughter languages have generalized different forms from different

¹⁴ Alternatively, since Karnic languages with deictic marking vary in the relative order of case marker and deictic marking, with stem–case–deictic and stem–deictic–case both appearing in the languages, it is possible that the Pitta-Pitta and Wangkajutjuru local cases reflect the stem–deictic–case ordering, and not the cliticization of a separate inflecting particle.

conjugations. Secondly, the tense-aspect systems of the languages differ somewhat from area to area, so it is not always certain whether similar forms with different meanings are in fact cognate. Thirdly, there appears to have been contamination from the deictic system, so that some tenses are ‘reinforced’ with a particle (again, which are not always cognate between languages). Therefore until reconstructions can be surer, I have chosen not to present verbal reconstructions here. This should not affect the argument, however, since the case reconstructions are solid.

One form that does have an etymology is the purposive *-linga*. It appears to be comprised historically of the nominalizer *-li* and the purposive case *-nga*,¹⁵ although synchronically there is little direct evidence for the segmentation. The nominal status of *li*-marked verbs can be seen in the following example:

(20)

- a. *Ngapu nhuwa-i* *watyi-linga, nhuwa-i* *thima-linga*.
 water 3SG.NFUT.NOM-UNDEF wash-PURP 3SG.NFUT.NOM-UNDEF drink-PURP
 ‘This water’s for washing, this one’s for drinking.’ (Blake & Breen 1971: 54)
- b. *Kathi nhuwa-ka* *piyawarli-nga*.
 meat 3SG.NFUT.NOM-UNDEF dog-PURP.
 ‘This meat is for the dog.’ (Blake 1979c:197)

The incorporation of the purposive case in the *-linga* construction is parallel to the use of the purposive with nouns, as seen by the parallel between (20a) and (20b).

3.5. Summary of Reconstructions

Table 4 below shows the shift in case forms and functions between the stage of Proto Karnic and the modern languages.

4. Syntactic Reconstruction of Tense-based Case Marking

The future clause case marking shows several differences from other Pama-Nyungan languages: *-ngu* marking both transitive and intransitive subjects; *-ku* marking direct rather than indirect objects; and null marking on the verb. The non-future clauses, on the other hand, show expected case marking: ergative/nominative/accusative, with suffixes which have widespread cognates in Karnic and other Pama-Nyungan languages. It is interesting to consider how such a tense-marking pattern could have arisen (and discussion will focus on deriving the pattern of marking in future contexts, since non-future clauses show a pattern most likely retained from previous stages of the language).

There are three plausible origins of this construction. The first is ‘insubordination’ [the terminology is from Evans (1995)]. This is the process whereby a subordinate clause comes over time to be used as a matrix clause. Subordinative morphology

¹⁵ The sources vary as to the transcription of the gerund formative/nominalizer as *-li* or *-lhi*. The potential *-lhi(R)a* may also contain this morphology, although I cannot identify the element *-Ra* at present.

Table 4. Case ‘shuffling’ in earlier stages of Pitta-Pitta and Wangkujutjuru

	Erg/Inst	Nom	Acc	Dat	Purp	Loc
Proto-Karnic (Bowern 1998)						
All tenses	*-lu -ngu	*-ø	*-nha	*-ku		*-nga
Pitta-Pitta (Roth 1897)						
Non-Future	-lu	-ø	-nha	-ku	-nga	-ina
Future	-ngu	-ngu	-ku	-ku	-nga	-ina
Pitta-Pitta (Blake & Breen 1971; Blake 1979)						
Non-Future	-lu	-ø	-nha	-ku	-nga	-ina
Future	-ngu	-ngu	-nha/-ku	-ku	-nga	-ina
Wangkajutjuru (Blake & Breen 1971)						
Non-Future	-lu	-ø	-nha	-ku	-nga	-na
Future	-ngu	-ngu	-nha	-ku	-nga	-na

remains, although the matrix clause does not. The second possible etymological source is the ‘less-affected object’ or ‘conative’ construction common in Australian languages, including Kalkatungu, Jalarnga and Warlpiri. The third is a reanalysed anti-passive, the solution suggested in passing by Hopper and Thomson (1980) in their discussions on this type of tense marking.

Each hypothesis will be evaluated in the following sections. Note that any hypothesis as to the origin of the future-marking has to account for four aspects of the construction:

1. the reflex of the Proto-Karnic ergative *-ngu as a future subject;
2. the reanalysis of the split between the ergative allomorphs *-lu and *-ngu;
3. the reflex of the Proto-Karnic dative *-ku as a future object;
4. the null exponence of future-marking on the verb.

4.1. Insubordination

One possible grammaticalization path is insubordination: i.e. that the marking arose in a parallel way to that in which tense-based case marking arose in Kayardild. In Kayardild the tense of a clause is marked not only on the verb (as a suffix), but also on all non-subject constituents. The case of non-subject arguments varies according to the tense and mood of the verb.¹⁶ The term ‘modal case’ was coined for this in Evans (1995). For example, the unmarked choice for the case for non-subject

¹⁶ Kayardild morphology is extremely complicated and the facts have been somewhat simplified here.

constituents in the past tense is ablative, but in the potential, the propriative is used. The verbal suffixes for tense play a somewhat secondary role to the modal case system, but tense marking on verbs does exist. In many instances, the suffixes are cognate with (or identical to) nominal/modal case suffixes. See Evans (1995: 253ff) for discussion. A few examples are given in (21) below:

- (21) (Kayardild)
- a. *Ngada dathin-kiring-ku kamarr-iring-ku.*
 1SG.NOM that-ALL-M.PROP stone-ALL-M.PROP
 'I will (go) to that stone.' (Evans 1995: 403, ex. 10-7)
 - b. *Ngada kurri-nangku mala-wu (balambi-wu).*
 1SG.NOM see-NEG.POT sea-M.PROP morrow-M.PROP
 'I won't (be able to) see the sea (tomorrow).' (Evans 1995: 404, ex. 10-12)
 - c. *Ngada kurri-nangku mala-y (barruntha-y).*
 1SG.NOM see-NEG.POT sea-M.LOC yesterday-M.LOC
 'I could not see the sea (yesterday).' (Evans 1995: 404, ex. 10-13)

The tense marking on nouns here is superficially similar to the strategy used in Pitta-Pitta and Wangkajutjuru, although recall that in the latter languages, the subject cases (ergative and nominative) have future allomorphs as well as the object cases, whereas in Kayardild it is normally the internal arguments and adjuncts that have case assigned on the basis of tense, not the external argument of the verb.

It is fairly clear that in Kayardild the tense marking on nouns is derived historically from marking all arguments in a subordinate clause with case suffixes. That is, the clauses were originally nominalized subordinate clauses. The case used depended on the semantics of the clause; that is, ablative case was used for clauses denoting cause, dative for purpose clauses, and the like. One finds such subordinative marking in many Australian languages, including Pitta-Pitta and Wangkajutjuru. Over time, Evans argues, the matrix clause was omitted or implied, but the nominalized clause (and the nominal marking) remained as the sole indication of the meaning of the implied matrix clause. Given the case marking patterns, as well as the fact that many verbal tense suffixes in Kayardild consist of an element and a form homophonous with the modal case marker used in that tense, this is certainly the correct historical explanation for the Kayardild's synchronic tense marking.

Now, applying this type of change to the situation in Pitta-Pitta and Wangkajutjuru would imply that at one stage there were nominalized clauses which took the dative case after some matrix verbs. These matrix verbs probably denoted some potential or purposive meaning, such as *it's possible that ...*; *I think that ... (might happen)*; *I intend to* At some stage the matrix clause was lost, leaving the modal case and synchronic tense-determined case allomorphy.

I do not favour this solution for Pitta-Pitta and Wangkajutjuru, although it is obviously correct for Kayardild. There are several reasons for this. First, in both Pitta-Pitta and Wangkajutjuru, the suffix on the verb in the future tense is a null morpheme. If Pitta-Pitta's tense based case marking came from insubordination, we would expect an overt suffix on the verb, some relic of the nominalizer or (if there was no nominalizer) the case marker which would have also appeared on the verb,

as it does in modern Pitta-Pitta. We would have to assume that such marking was lost from the verb. Alternatively, we would have to assume that future tense-based case marking was generalized from a context other than the future tense (for example, the purposive *-linga*). This solves the problem of the lack of an overt morpheme on the verb (indeed *-linga* probably shows a relic of purposive case-marking in the *-nga* portion of the suffix). It raises other problems, however. *-nga*, the modern purposive, is historically from the locative and *-ku* from the old dative/purposive. Now, *-nga* cannot shift to purposive before *-ku* ceases to be used as the purposive. Since the semantics of *-linga* (verbal purposive) are dependent on the purposive reading of *-nga*, *-linga* cannot be insubordinated before *-ku* is reanalysed. Therefore this construction is rather unlikely to be the locus of spread of *-ku* marking futures. It is much simpler to take the main clause future clauses as the origin of the construction and have it spread to clauses marked by *-linga* after *-nga* has taken over the functions of the purposive.

Therefore it seems that arguments regarding insubordination as the donor construction are not tenable. The historical changes required to transform one into the other are convoluted, and include the loss of all verbal suffixes and the separate innovation of tense-based subject marking. Point 3 is explained by this scenario, but points 1 and 2 are not, and point 4 presents problems.

4.2. *Anti-passives*

The anti-passive is a valency-reducing derivation which demotes direct object arguments to oblique noun phrases and renders transitive verbs intransitive [see, for example, Blake (1979b) for Kalkatungu and Terrill (1997) for a survey of anti-passives in Australia]. The typical morphological expression is nominative case on the subject, dative or locative case on the demoted 'object' and an affix to the verb.

Hopper and Thomson (1980: 273–274) tie the tense-determined case marking of Pitta-Pitta to such a construction; they suggest that Pitta-Pitta's case split is in origin an active/anti-passive split. This motivates the use of *-ku* on objects (point 3 above). They do not discuss the other uses of *-ku* but one would assume that they would argue that the use of *-ku* with other clause types (such as imperatives) is a later innovation. Dixon (2002: 542–543) makes the same claim.

There are two reasons why this argument is implausible. Firstly, the anti-passive construction in languages in which it occurs in Australia is marked with an overt suffix on the verb [recall example (14) above]. Neither Pitta-Pitta nor Wangka-jutjuru shows an overt verbal suffix in the future tense (although they do in the anti-passive, where the suffix is *-h*); this fact is not explained by the anti-passive argument and indeed counts against it.

Secondly, and more importantly, however, the origin of the most common future subject suffix appears without much doubt to be the ergative; yet the anti-passive is the one construction where an ergative suffix, by the definition of the construction type, would be excluded! The anti-passive has a detransitivizing function; those languages which have such a construction without exception mark the subjects of anti-passive verbs as nominative. Those who seek to explain the origin of Pitta-

Pitta's future tense marking in an anti-passive construction must find a way to introduce an allomorph of the ergative into a position where we do not expect ergative marking. Such a generalization would have to happen later and be independent of the creation of future-marking on objects.

Thus I do not consider the anti-passive to be a viable 'donor' construction for tense-based case marking in Pitta-Pitta. Point 3 is satisfied, however points 1 and 4 present difficulties for this scenario, and point 2 is not addressed.

4.3. *Less-affected Objects*

It is quite common in Australian languages for 'less affected arguments' [using terminology from Hopper and Thomson (1980)] of transitive verbs, such as intentional objects, unattained goals, and the like, to be marked differently from 'more affected arguments'. In Kalkatungu, for example, clauses with imperfective aspect mark direct objects with dative case, as shown in (22):

(22) (Kalkatungu)

- a. *Thuku-yu yuru-ø ityai-nha.*
 dog-ERG man-ABS bite-PST
 'The dog bit the man.'
- b. *Thuku-ø nguli ityai ruupu-wu ai ingka.*
 dog-NOM CONT bite rope-DAT COMP.3S go
 'The dog keeps chewing at the rope to get away.' (Blake 1977: 79)

In (22a) the subject is ergative-marked and the object, 'man', is in the absolutive. In (22b), however, the subject is nominative and the object is dative.

A link is noted in Hopper and Thomson (1980) between the future tense and 'reduced transitivity' (future or irrealis is one of the factors on their transitivity hierarchy). Now, since we can trace the Pitta-Pitta and Wangkajutjuru forms marking future tense on nouns to an allomorph of the Proto Karnic ergative and the Proto Karnic dative, we should look in other related Australian languages for a construction using these two cases which marks unattained goals, less affected objects, or reduced transitivity.

We find such a construction in Warlpiri. The object is demoted from absolutive to dative, and ergative case marking remains on the subject [as described for example in Nash (1986: 198ff)]. This occurs with certain verbs, such as *warri-rni* 'seek' and the preverb *wapal*, 'in search of' (possibly cognate with the Pitta-Pitta verb *wapa-* 'look for', which as noted above always takes *-ku*), as well as some verbs where the object is also the goal of motion [for details see Nash (1986), quoting Hale (1978: 47)]. Example (23) below gives an example of the verb *paka* 'hit' with ergative and absolutive, and ergative and dative case frames.

(23) (Warlpiri)

- a. *Wati-ngki marlu-ø luwa-rnu.*
 man-ERG kangaroo-ABS shoot-PST.
 'The man shot the kangaroo.'

- b. *Wati-ngki-rla-jinta marlu-ku luwa-rnu.*
 man-ERG-3DAT-DAT kangaroo-DAT shoot-PST.
 'The man shot at the kangaroo.' (Nash 1986: 198)

I suggest an origin of the Pitta-Pitta marking in this type of construction. The path of semantic change from intensional object in the present (that is, a 'goal') to an object in the future (which is also a goal not yet attained) is a plausible one. The shift is in the focus of the construction from the unattained nature of the object, with action taking place in the present (e.g. 'I am striking at the dog, *but I am not currently actually hitting it*') to the possibility that the object will be attained in the future [e.g. '*I am not currently actually hitting the dog*, but in the future I might/will strike (at) it']. Note also the association in Hopper and Thomson (1980) between transitivity, futurity and imperfectivity; that is, that like imperfective clauses, future and potential clauses have by implication unattained goals or objects. I am arguing here that it is this association that seems to have been exploited in the reanalysis of ergative-dative case marking in Pitta-Pitta and Wangkajutjuru. In this scenario, the dative marking (point 3) is accounted for. We have a source for the ergative (point 1) as well, since it follows from the case frame of the donor construction. Point 4 is accounted for as well; we require no additional verbal morphology to signal the construction.

The one aspect of future case-marking that this analysis does not account for is point 2, the extension of the morpheme *-ngu* from an ergative marker to a combined ergative/nominative (that is, 'subject') case (an aspect of the grammaticalization of future marking in Pitta-Pitta that none of the other scenarios account for either). How did *-ngu* come to be extended from transitive clauses to intransitive clauses as well?

One source could be that at some point *-ngu* was felt to mark 'future subject' rather than 'future transitive subject', that is, that the primary function of core case marking was not to signal transitivity alternations, but tense alternations. Another possibility (which is also compatible with the previous explanation) is that the reduced transitivity of the future resulted in the demotion of the ergative marked argument to a 'subject' case, parallel to the pattern we find in Kalkatungu (compare for example Blake 1979a). Nominative/Dative case frames for reduced transitivity constructions are quite common and there may have been some areal pressure in that direction.

5. Conclusions

The synchronic conditions for marking tense on nominals have been briefly summarized. The dialects differ as to their treatment of tense marking and the last speakers were not consistent in their use of the separate future suffixes. Three possible etymologies of Pitta-Pitta's tense-determined case marking have been discussed. Of these, the anti-passive is unlikely because it required us to assume the marking of transitive subjects as ergative in an environment where they would not have been marked with such a case. The analysis of the construction as a reanalysed present

'conative' has the advantage that there are examples of this construction in neighbouring and related languages which match the Pitta-Pitta construction in almost every detail. It fits the morphology; we do not have to explain the disappearance of verb morphology or the presence of historically ergative morphology (both of which are serious problems for a reconstruction of the future tense marking as the reanalysis of an anti-passive).

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